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Windows PowerShell

(.venv) PS C:\Users\Sanjana CA\OneDrive\Documents\CynarisInternship\W11D4_RAG_Evaluation> python app\rag_evaluation.py
Loading healthcare dataset...
Loaded documents: 1
Created chunks: 32
Creating ChromaDB...
ChromaDB created successfully.

=====
RAG TEST
=====

Question:
What are common symptoms of diabetes?

Answer:
Common symptoms of diabetes include increased thirst, frequent urination, increased hunger, fatigue, and blurred vision.

Retrieved Context:

--- Context 1 ---
or does not produce enough insulin. Common symptoms of diabetes can include increased thirst, frequent urination, increased hunger, fatigue, and blurred vision. Diabetes can be managed through healthy eating, physical activity, medication when prescribed, and regular monitoring of blood glucose

=====
RAG pipeline test completed.
=====
(.venv) PS C:\Users\Sanjana CA\OneDrive\Documents\CynarisInternship\W11D4_RAG_Evaluation> python app\optimized_rag.py
Loading healthcare dataset...
Loaded documents: 1
Created chunks: 32
Creating optimized ChromaDB...
Optimized ChromaDB created successfully.

=====
OPTIMIZED RAG TEST
=====

Question:
What are common symptoms of diabetes?

Answer:
Common symptoms of diabetes include increased thirst, frequent urination, increased hunger, fatigue, and blurred vision.
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Number of retrieved chunks: 3

Retrieved Context:

--- Context 1 ---
or does not produce enough insulin. Common symptoms of diabetes can include increased thirst, frequent urination, increased hunger, fatigue, and blurred vision. Diabetes can be managed through healthy eating, physical activity, medication when prescribed, and regular monitoring of blood glucose

--- Context 2 ---
Diabetes is a chronic condition in which blood glucose levels remain higher than normal. Type 1 diabetes occurs when the immune system attacks insulin-producing cells in the pancreas. Type 2 diabetes occurs when the body does not use insulin effectively or does not produce enough insulin. Common

--- Context 3 ---
Healthcare Knowledge Base

1. Diabetes

=====
OPTIMIZED RAG PIPELINE TEST COMPLETED
=====
(.venv) PS C:\Users\Sanjana CA\OneDrive\Documents\CynarisInternship\W11D4_RAG_Evaluation> Get-Content output\evaluation_dataset.json
[
  {
    "question": "What are common symptoms of diabetes?",
    "answer": "Common symptoms of diabetes can include increased thirst, frequent urination, increased hunger, fatigue, and blurred vision. Diabetes can be managed through healthy eating, physical activity, medication when prescribed, and regular monitoring of blood glucose",
    "contexts": [
      "or does not produce enough insulin. Common symptoms of diabetes can include increased thirst, frequent urination, increased hunger, fatigue, and blurred vision. Diabetes can be managed through healthy eating, physical activity, medication when prescribed, and regular monitoring of blood glucose"
    ],
    "ground_truth": "Common symptoms of diabetes include increased thirst, frequent urination, increased hunger, fatigue, and blurred vision."
  },
  {
    "question": "What is hypertension?",
    "answer": "Hypertension.",
    "contexts": [
      "2. Hypertension"
    ],
    "ground_truth": "Hypertension is a condition in which the force of blood against artery walls remains consistently elevated."
  },
  {
    "question": "What are common symptoms of asthma?",
    "answer": "Common symptoms of asthma include wheezing, coughing, chest tightness, and shortness of breath.",
    "contexts": [
      "Asthma is a chronic respiratory condition that causes inflammation and narrowing of the airways. Symptoms may include wheezing, coughing, chest tight"
    ]
  }
]
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ness, and shortness of breath. Asthma symptoms can be triggered by allergens, respiratory infections, exercise, smoke, and other environmental"
},
"ground_truth": "Common symptoms of asthma include wheezing, coughing, chest tightness, and shortness of breath."
},
{
"question": "What is one common cause of anemia?",
"answer": "Iron deficiency is one common cause of anemia.",
"contexts": [
"The common cold is a viral infection that usually affects the upper respiratory tract. Symptoms can include a runny nose, sore throat, coughing, sneezing, and mild fatigue. Most common colds improve with rest, fluids, and supportive care. Antibiotics do not treat viral infections such as the common cold."
],
"ground_truth": "Iron deficiency is one common cause of anemia."
},
{
"question": "Are antibiotics effective against the common cold?",
"answer": "No, antibiotics are not effective against the common cold.",
"contexts": [
"The common cold is a viral infection that usually affects the upper respiratory tract. Symptoms can include a runny nose, sore throat, coughing, sneezing, and mild fatigue. Most common colds improve with rest, fluids, and supportive care. Antibiotics do not treat viral infections such as the common cold."
],
"ground_truth": "No. Antibiotics do not treat viral infections such as the common cold."
},
{
"question": "What is influenza?",
"answer": "Influenza.",
"contexts": [
"6. Influenza"
],
"ground_truth": "Influenza, or flu, is a contagious respiratory illness caused by influenza viruses."
},
{
"question": "What are some risk factors for heart disease?",
"answer": "According to the provided context, some risk factors for heart disease include:\n\n1. High blood pressure\n2. High cholesterol\n3. Smoking\n4. Diabetes\n5. Physical inactivity (implied by the mention of \"physical inactivity\" as a separate point, but not explicitly listed as a risk factor)",
"contexts": [
"Heart disease refers to several conditions affecting the heart and blood vessels. Coronary artery disease is one common form and occurs when the arteries supplying the heart become narrowed or blocked. Risk factors can include high blood pressure, high cholesterol, smoking, diabetes, physical inactivity, and family history."
],
"ground_truth": "Risk factors can include high blood pressure, high cholesterol, smoking, diabetes, physical inactivity, and family history."
},
{
"question": "What do the kidneys do?",
"answer": "The kidneys filter waste and excess fluid from the blood and help regulate fluid, electrolytes, and blood pressure.",
"contexts": [
"The kidneys filter waste and excess fluid from the blood and help regulate fluid, electrolytes, and blood pressure. Chronic kidney disease involves a gradual loss of kidney function over time. Diabetes and hypertension are important risk factors for chronic kidney disease. Regular medical"
],
"ground_truth": "The kidneys filter waste and excess fluid from the blood and help regulate fluid, electrolytes, and blood pressure."
},
{
"question": "What does a balanced diet provide?",
"answer": "A balanced diet provides carbohydrates, proteins, fats, vitamins, minerals, fiber, and water.",
"contexts": [
"A balanced diet provides the body with carbohydrates, proteins, fats, vitamins, minerals, fiber, and water. Fruits, vegetables, whole grains, legumes, nuts, and other nutrient-dense foods can contribute to a healthy diet. Individual nutritional needs vary depending on age, activity level, health"
],
"ground_truth": "A balanced diet provides carbohydrates, proteins, fats, vitamins, minerals, fiber, and water."
},
{
"question": "What is preventive healthcare?",
"answer": "Preventive healthcare refers to measures taken to prevent or reduce the risk of illness, injury, or disease.",
"contexts": [
"10. Preventive Healthcare"
],
"ground_truth": "Preventive healthcare focuses on reducing the risk of disease and detecting health problems early."
},
{
}
}
}

(.venv) PS C:\Users\Sanjana CA\OneDrive\Documents\CynarisInternship\W11D4_RAG_Evaluation> Get-Content output\RAGAS_EVALUATION_REPORT.md
\# W11D4 RAG Evaluation Report

\## Project

W11D4: RAG Evaluation with Ragas & Metrics & Benchmarks

\## Dataset

Healthcare knowledge dataset containing 10 topics:

1\ Diabetes
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"The kidneys filter waste and excess fluid from the blood and help regulate fluid, electrolytes, and blood pressure. Chronic kidney disease involves a gradual loss of kidney function over time. Diabetes and hypertension are important risk factors for chronic kidney disease. Regular medical"
},
"ground_truth": "The kidneys filter waste and excess fluid from the blood and help regulate fluid, electrolytes, and blood pressure."
},
{
"question": "What does a balanced diet provide?",
"answer": "A balanced diet provides carbohydrates, proteins, fats, vitamins, minerals, fiber, and water.",
"contexts": [
"A balanced diet provides the body with carbohydrates, proteins, fats, vitamins, minerals, fiber, and water. Fruits, vegetables, whole grains, legumes, nuts, and other nutrient-dense foods can contribute to a healthy diet. Individual nutritional needs vary depending on age, activity level, health"
],
"ground_truth": "A balanced diet provides carbohydrates, proteins, fats, vitamins, minerals, fiber, and water."
},
{
"question": "What is preventive healthcare?",
"answer": "Preventive healthcare refers to measures taken to prevent or reduce the risk of illness, injury, or disease.",
"contexts": [
"10. Preventive Healthcare"
],
"ground_truth": "Preventive healthcare focuses on reducing the risk of disease and detecting health problems early."
},
{
}
}

(.venv) PS C:\Users\Sanjana CA\OneDrive\Documents\CynarisInternship\W11D4_RAG_Evaluation> Get-Content output\RAGAS_EVALUATION_REPORT.md
\# W11D4 RAG Evaluation Report

\## Project

W11D4: RAG Evaluation with Ragas & Metrics & Benchmarks

\## Dataset

Healthcare knowledge dataset containing 10 topics:

1\ Diabetes
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2\. Hypertension
3\. Asthma
4\. Anemia
5\. Common Cold
6\. Influenza
7\. Heart Disease
8\. Kidney Health
9\. Nutrition
10\. Preventive Healthcare

\## Evaluation Dataset

10 question-answer pairs were generated from the healthcare knowledge dataset and stored in:

'output/evaluation\_dataset.json'

\## Baseline RAG Configuration

\~ Chunk size: 300
\~ Chunk overlap: 50
\~ Retrieval k: 1
\~ Embedding model: 'nomic-embed-text:latest'
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\~ Embedding model: 'nomic-embed-text:latest'
\~ Generation model: 'llama3.2:3b'
\~ Vector store: ChromaDB

\## Optimized RAG Configuration

The retrieval configuration was optimized by increasing the number of retrieved chunks:

\~ Chunk size: 300
\~ Chunk overlap: 50
\~ Retrieval k: 3
\~ Embedding model: 'nomic-embed-text:latest'
\~ Generation model: 'llama3.2:3b'
\~ Vector store: ChromaDB

The optimized pipeline successfully retrieved 3 chunks for each query and generated context-based answers.

\## Ragas Metrics

The following Ragas metrics were attempted:

\~ Faithfulness
\~ Answer Relevancy
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\~ Context Precision
\~ Context Recall

\## Ragas Evaluation Attempt

Ragas 0.4.3 was used with a local Ollama model as the evaluation judge.

The evaluation was attempted using:

\~ 'llama3.2:3b'
\~ 'qwen2.5:3b'

Both local models produced timeout errors and invalid structured JSON responses during Ragas metric evaluation.

A single-question diagnostic test was also performed using Faithfulness. The evaluation timed out after approximately 3 minutes and returned:

{'faithfulness': nan}

The full 10-question optimized evaluation completed its 40 metric jobs, but the local judge generated multiple 'OutputParserException' and 'TimeoutError' failures. Therefore, valid numerical scores could not be reliably produced for the complete evaluation.

\## Important Evaluation Note
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No Ragas scores were fabricated or manually assigned.

The 'null' values in the generated JSON result files indicate that the local Ragas judge evaluation did not produce reliable numerical results.

This limitation is specific to the local LLM-as-judge evaluation setup. The RAG retrieval and answer-generation pipeline itself executed successfully.

\## Optimization Performed

The retrieval parameter was changed from:

'k = 1'

to:

'k = 3'

The optimized pipeline was rebuilt with a clean ChromaDB collection to avoid duplicate vector entries.

The optimized RAG pipeline successfully retrieved 3 context chunks and generated answers from the retrieved healthcare information.

\## Baseline Ragas Console Evidence
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During the initial baseline evaluation, Ragas displayed the following aggregate values:

\~ Faithfulness: 1.0000
\~ Answer Relevancy: 0.8528
\~ Context Precision: 0.0000
\~ Context Recall: 0.9286

These values were displayed by the Ragas evaluation process, but the original result-extraction code did not preserve them correctly in the JSON output. Because the complete evaluation also contained failed jobs, these values are retained as console evidence rather than treated as a complete reliable benchmark.

\## Conclusion

The W11D4 RAG evaluation workflow was implemented successfully, including dataset preparation, baseline retrieval, optimized retrieval, ChromaDB indexing, Ollama generation, and Ragas evaluation attempts.

The main limitation encountered was the inability of the available local 3B Ollama models to consistently satisfy Ragas' structured LLM-judge output requirements within the available execution time.

The optimization experiment was implemented by increasing retrieval k from 1 to 3, and the optimized RAG pipeline was verified independently.

(.venv) PS C:\Users\Sanjana CA\OneDrive\Documents\CynarisInternship\W11D4_RAG_Evaluation> Get-Content output\SELF_REVIEW_CHECKLIST.md
\# W11D4 Self-Review Checklist

\## RAG Pipeline
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\## RAG Pipeline
\~ \[x] Healthcare dataset prepared
\~ \[x] 10 evaluation Q&A pairs generated
\~ \[x] LangChain RAG pipeline implemented
\~ \[x] ChromaDB vector store configured
\~ \[x] Baseline retrieval implemented with k=1
\~ \[x] Optimized retrieval implemented with k=3
\~ \[x] Optimized ChromaDB rebuilt without duplicate entries
\~ \[x] Ollama embeddings configured
\~ \[x] Ollama generation model configured

\## Ragas Evaluation
\~ \[x] Ragas installed and imported successfully
\~ \[x] Faithfulness evaluation attempted
\~ \[x] Answer Relevancy evaluation attempted
\~ \[x] Context Precision evaluation attempted
\~ \[x] Context Recall evaluation attempted
\~ \[x] Baseline evaluation executed
\~ \[x] Optimized evaluation executed
\~ \[x] Local LLM evaluation limitation documented
\~ \[x] No unsupported or fabricated scores used

\## Optimization
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\\## Optimization
\\- \\[x] Baseline retrieval k=1 documented
\\- \\[x] Optimized retrieval k=3 documented
\\- \\[x] Optimized pipeline verified successfully
\\- \\[x] Retrieved context count verified as 3

\\## Documentation
\\- \\[x] Ragas evaluation report created
\\- \\[x] Evaluation dataset saved
\\- \\[x] Baseline result file saved
\\- \\[x] Optimized result file saved
\\- \\[x] Evaluation limitation documented

\\## Code Quality
\\- \\[x] Code organized inside app/
\\- \\[x] Data stored inside data/
\\- \\[x] Evaluation outputs stored inside output/
\\- \\[x] Python virtual environment used
\\- \\[x] Local Ollama models used for RAG generation and evaluation attempts

\\## Final Status
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\\- \\[x] Retrieved context count verified as 3

\\## Documentation
\\- \\[x] Ragas evaluation report created
\\- \\[x] Evaluation dataset saved
\\- \\[x] Baseline result file saved
\\- \\[x] Optimized result file saved
\\- \\[x] Evaluation limitation documented

\\## Code Quality
\\- \\[x] Code organized inside app/
\\- \\[x] Data stored inside data/
\\- \\[x] Evaluation outputs stored inside output/
\\- \\[x] Python virtual environment used
\\- \\[x] Local Ollama models used for RAG generation and evaluation attempts

\\## Final Status

W11D4 implementation completed.

The RAG pipeline and retrieval optimization were successfully implemented and verified. Ragas LLM-as-judge evaluation was attempted for all required metrics, but reliable numerical scores could not be obtained because the available local 3B models produced timeouts and invalid structured outputs. This limitation has been documented instead of fabricating benchmark results.

(.venv) PS C:\\Users\\Sanjana CA\\OneDrive\\Documents\\CynarisInternship\\W11D4_RAG_Evaluation>
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about the kidneys' function, which are reflected in the answer.  
For troubleshooting, visit: https://docs.langchain.com/oss/python/langchain/errors/OUTPUT_PARSING_FAILURE }  
Evaluating: 80%|███████████ | 32/40 [1:01:03<06:49, 51.18s/it]Exception raised in Job[  
32]: TimeoutError() | 0/1 [00:00<?, ?it/s]  
Evaluating: 82%|███████████ | 33/40 [1:04:03<10:28, 89.83s/it]Exception raised in Job[  
33]: OutputParserException(Invalid json output: Given the input: | 0/1 [00:00<?, ?it/s]  
{  
    "response": "A balanced diet provides carbohydrates, proteins, fats, vitamins, minerals, fiber, and water."  
}  
  
The generated question is:  
"What are the components of a balanced diet?"  
  
The answer is committal, as it provides specific information about the components of a balanced diet. Therefore, the output in JSON format is:  
  
{  
    "question": "What are the components of a balanced diet?",  
    "noncommittal": 0  
}  
  
For troubleshooting, visit: https://docs.langchain.com/oss/python/langchain/errors/OUTPUT_PARSING_FAILURE }  
Evaluating: 85%|███████████ | 34/40 [1:05:43<09:15, 92.64s/it]Exception raised in Job[  
34]: OutputParserException(Invalid json output: The provided context was indeed useful in arriving at the given answer. The context includes key information  
about a balanced diet, which are reflected in the answer.  
For troubleshooting, visit: https://docs.langchain.com/oss/python/langchain/errors/OUTPUT_PARSING_FAILURE }  
Evaluating: 92%|███████████ | 37/40 [1:07:34<02:46, 55.44s/it]Exception raised in Job[  
37]: OutputParserException(Invalid json output: Given the input: | 0/1 [00:00<?, ?it/s]  
{  
    "response": "Preventive healthcare refers to measures taken to prevent or reduce the risk of illness, injury, or disease."  
}  
  
We can generate a question and identify if the answer is noncommittal as follows:  
  
Question: What does preventive healthcare refer to?  
  
Answer is committal, so noncommittal answer is 0.  
  
Output in JSON format:  
  
{  
    "question": "What does preventive healthcare refer to?",  
    "noncommittal": 0  
}  
  
For troubleshooting, visit: https://docs.langchain.com/oss/python/langchain/errors/OUTPUT_PARSING_FAILURE }  
Evaluating: 100%|███████████ | 40/40 [1:09:08<00:00, 103.70s/it]
```

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Taskbar icons: Weather (29°), Start menu, Search, File Explorer, Edge, VS Code, Teams, Outlook, Mail, Chrome, Task View, System tray (Network, Volume, Power, Date/Time).